CPSC 368 · GROUP 2 · PHASE 6

Lights, Camera, Correlation

How production team characteristics shape movie ratings and career outcomes

A study of IMDb, MovieLens & TMDb

Introduction

BACKGROUND

The film industry runs on collaboration — between actors, directors, writers, and studios.

Yet there is little quantitative evidence showing how these interactions actually shape a film's success.

We ask whether common beliefs in the entertainment industry hold up against the data.

OUR THEME

How do the characteristics of film production teams influence movie ratings and career outcomes across different genres?

Ratings · Careers · Genres

Research Questions

THREE ANGLES ON ONE THEME

01 Star Power

Do appearances in highly-rated films translate into more future lead roles?

02 Creative Chemistry

Does repeated director–writer collaboration improve movie ratings across genres?

03 Cast Composition

How does the gender ratio of the principal cast affect average ratings?

Why It Matters

WHO CARES ABOUT THE ANSWERS

1

Studios & Producers

Smarter casting and creative partnership decisions

2

Casting Directors

Evaluating talent and long-term potential

3

Investors

Assessing risk and predictors of quality

4

Aspiring Artists

Understanding what drives career advancement

Data Sources

THREE COMPLEMENTARY DATASETS

IMDb

Non-Commercial Datasets

Titles, crew, cast, ratings — our primary source for film metadata.

MovieLens

GroupLens Research

The "Rosetta Stone" — its links.csv joins IMDb and TMDb IDs.

TMDb

Top 10,000 on Kaggle

Popularity metrics that help surface top-performing movies.

Joined on shared identifiers → one unified dataset for all three research questions

Data Cleaning

FROM RAW FILES TO ANALYSIS-READY DATA

1

Load Smart

First 50K rows loaded to protect system RAM

2

Drop Nulls

Removed rows with missing values to stabilize queries

3

Fix IDs

Added "tt" prefix to MovieLens IDs for clean joins

4

Standardize

Lowercased strings; converted pipe-separated genres to arrays

5

Sample Down

Randomly sampled the final set to 2,000 rows for efficiency

EDA HIGHLIGHT

Balanced casts were rare. Male-dominated casts dominated — but female-led action films stood out as outliers with unusually high ratings.

Methodology

AGGREGATION-BASED ANALYSIS IN BOTH SYSTEMS

We applied the same analytical logic in Oracle and MongoDB, built on four core statistical moves.

Counts

Roles per actor · collaborations per duo

Averages

Mean IMDb rating by group

Ratios

Male-to-total actor ratio in each cast

Grouping

Action · Romance · Comedy buckets

APPROACH

Oracle

SQL queries with multi-table joins and CTEs

MongoDB

Aggregation pipelines transforming embedded documents

Database Implementation

SAME DATA · TWO SCHEMAS

ORACLE

Relational · Normalized

- One table per cleaned CSV
- Movies, cast, crew, ratings kept separate
- Shared movie and person IDs link the tables
- Joined on demand during analysis

MONGODB

Document · Embedded

- Three collections: Movies, People, UserRatings
- Cast and crew embedded inside movie documents
- People kept separate — reused across many films
- Fewer joins; closer to how movie data is accessed

Comparison & Reflection

WHICH DATABASE WON?

OUR PICK

MongoDB

Movie data is naturally document-shaped. Cast, crew, genres, and ratings live together — most queries avoid joins entirely.

EASIER IN MONGODB

Cast-based analyses — gender ratios and role counts — just needed $unwind on embedded arrays.

EASIER IN ORACLE

Relational joins across separate datasets — SQL provides cleaner structure for combining normalized tables.

WHEN WE'D SWITCH

Oracle wins if the project shifts toward strict foreign-key integrity or heavy cross-entity joins rather than movie-centric lookups.

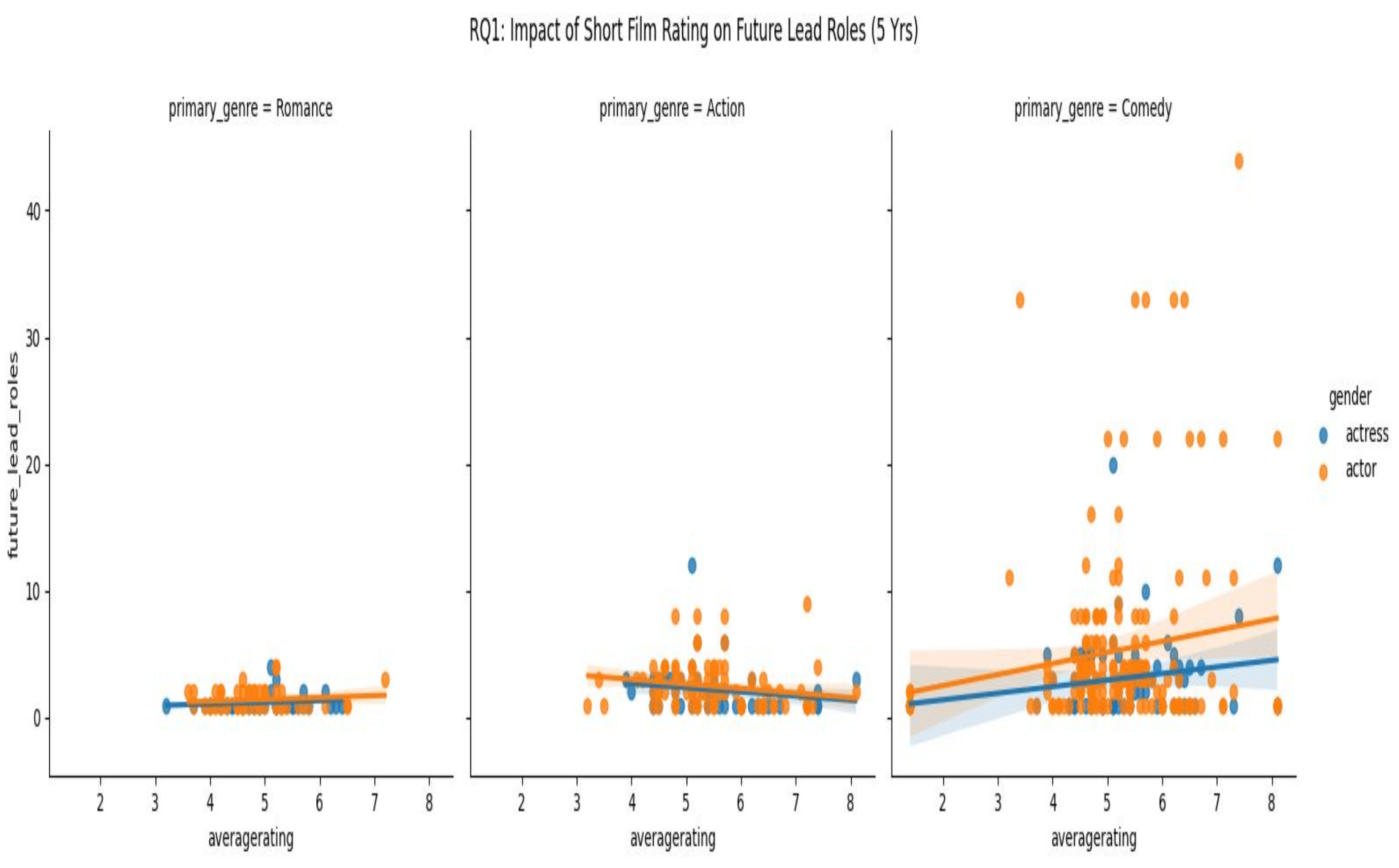
PART TWO

Results

Three research questions · three hypotheses · three surprises

RQ1 · Star Power

DO HIGH RATINGS CREATE FUTURE CAREERS?



Future lead roles vs. initial film rating, faceted by genre

HYPOTHESIS

High ratings lead to more lead roles in the next 5 years.

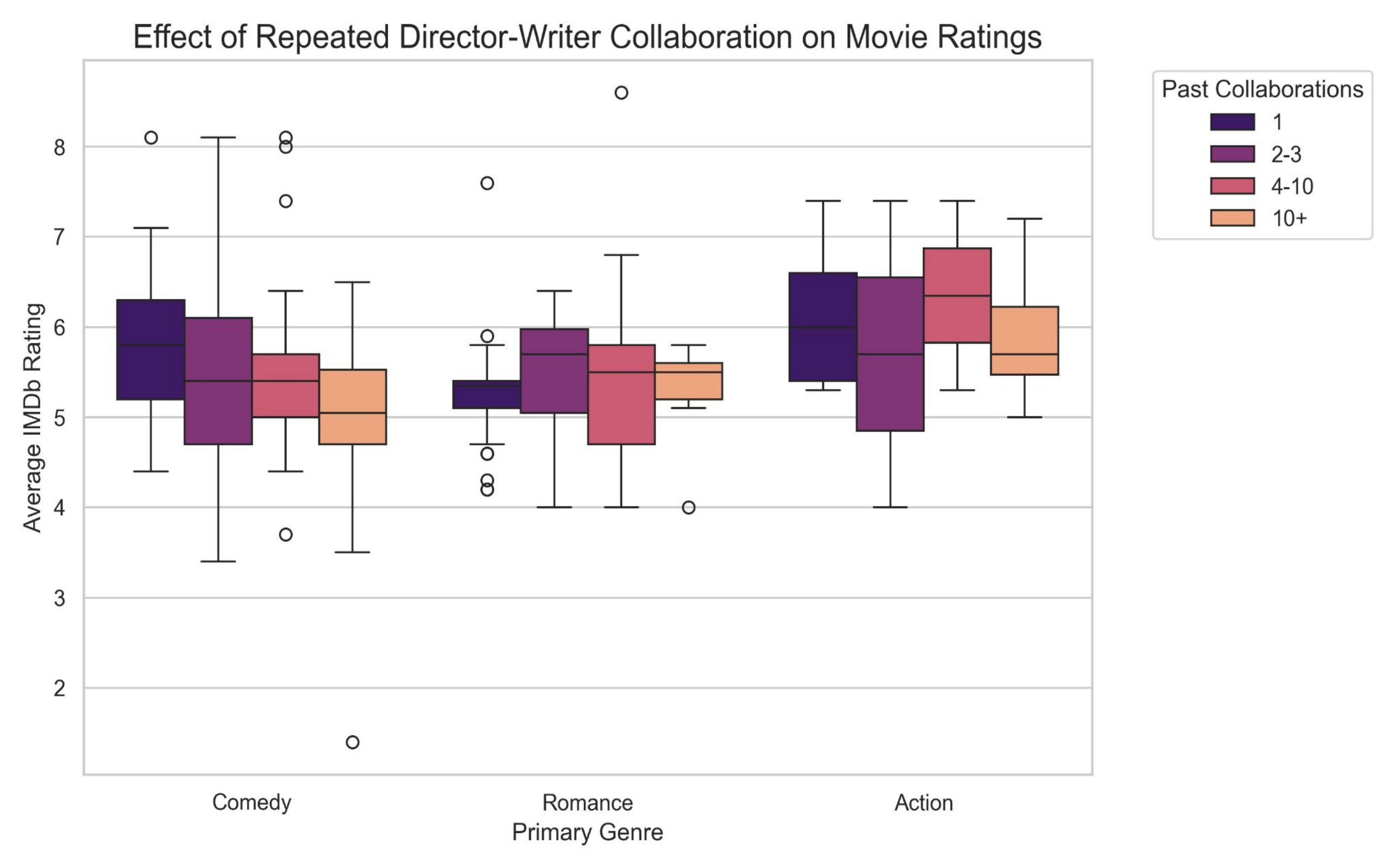
VERDICT · PARTIALLY CONTRADICTED

Comedy
Positive trend — higher ratings = more future roles, led by male actors.

Action & Romance
Flat regression lines — ratings do not predict role volume.

RQ2 · Creative Chemistry

DOES REPEATED COLLABORATION PAY OFF?



IMDb rating distributions by director–writer collaboration count

HYPOTHESIS

More collaborations build chemistry → higher ratings.

VERDICT · CONTRADICTED

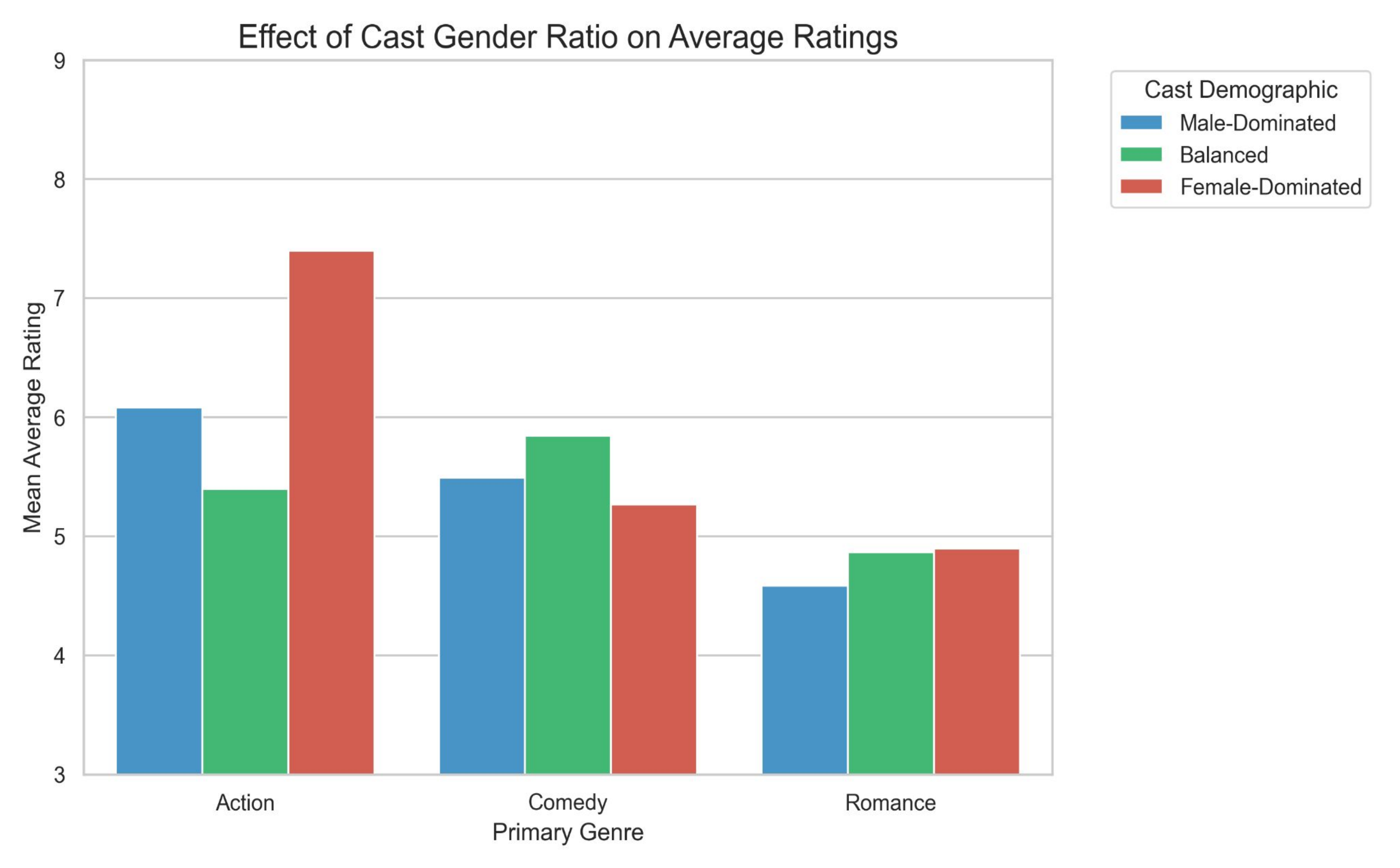
Comedy
Ratings steadily decline as collabs pile up.

Romance
Flat — repetition neither helps nor hurts.

Action
Sweet spot at 4–10 collabs; drops after.

RQ3 · Cast Composition

DOES GENDER BALANCE SHAPE RATINGS?



Mean IMDb rating by cast demographic and genre

HYPOTHESIS

Balanced casts earn the highest ratings across all genres.

VERDICT · PARTIALLY SUPPORTED

Comedy ✓
Balanced casts rated highest (~5.8).

Romance ~
Female-led and balanced tied; male-led lowest.

Action ✗
Female-dominated casts spiked to ~7.4.

Results at a Glance

THREE HYPOTHESES · THREE VERDICTS

	HYPOTHESIS	VERDICT
RQ1 Star Power	*High ratings → more future roles*	**Partially Contradicted** Only Comedy showed the expected trend; Action & Romance stayed flat.
RQ2 Creative Chemistry	*More collaborations → higher ratings*	**Contradicted** Fresh pairings often beat veteran duos; only Action had a brief sweet spot.
RQ3 Cast Composition	*Balanced casts → highest ratings*	**Partially Supported** True for Comedy; Romance was mixed; Action favored female-dominated casts.

Limitations

WHAT TO KEEP IN MIND WHEN READING OUR RESULTS

Sample Size	Capped at 2,000 rows to manage RAM — a tiny slice of millions of titles in the global industry.
Survivorship Bias	Dropping nulls favored modern, Western, big-budget films with complete metadata.
Validity Threat (RQ1)	No chronological filter on "future" roles — we captured correlation, not causation.
Omitted Variables	No budgets, marketing spend, or franchise IP — stronger drivers may be hidden behind our trends.

Future Work

QUESTIONS WE'D CHASE NEXT

IF WE HAD MORE TIME

- Scale to the full IMDb and MovieLens datasets on a distributed cloud database
- Integrate financial data — box office, budgets, marketing spend

OPEN QUESTIONS

The Female-Action Spike

Are female-dominated action films genuinely loved, or is it a tiny-sample mirage?

The Comedy Collaboration Penalty

Why does comedy alone punish repeated director–writer pairings?

THANK YOU

Questions?

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Data sources: IMDb · MovieLens · TMDb